Féidearthachtaí as Cuimse
Infinite Possibilities

AI: Theory to Practice

Modelling & Evaluation Presentation

Group 6:
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Odhran Chamberlain X00217322
Oleh Kravets X00219607

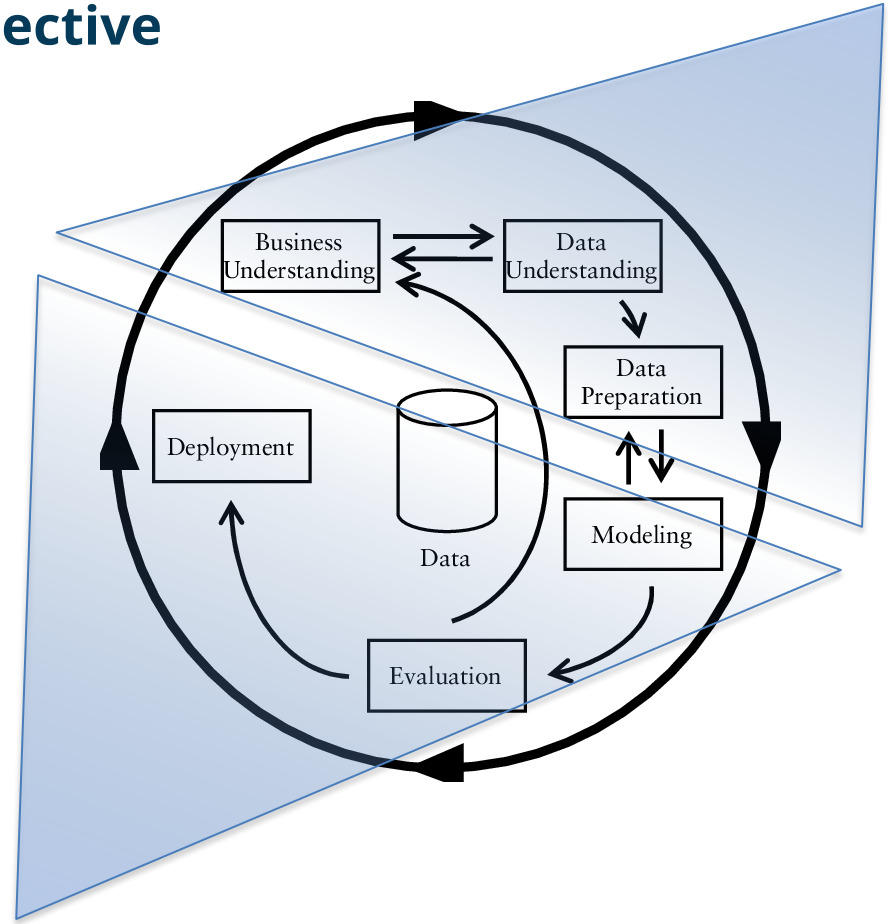
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Project Objective

CA1

In this assignment, we will work in a group of 3 to:

- Identify a publicly available dataset (≥ 1,000 rows, ≥ 8 columns).
- Define a clear supervised machine learning problem (classification or regression).
- Conduct an exploratory data analysis (EDA) (data understanding) and data preparation process to make the dataset model-ready.



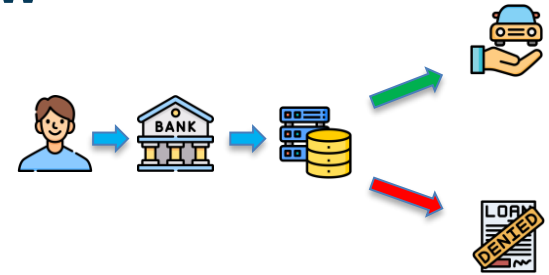
CA2

In this assignment, we will work in a group of 3 to:

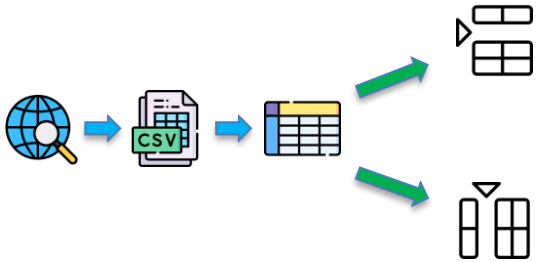
- Design and apply feature engineering, feature selection, and any necessary scaling or transformations.
- Build and tune at least four supervised ML models (three traditional and one simple deep neural network) using a correct ML workflow, including data splitting and handling issues such as class imbalance and overfitting.
- Evaluate the models using appropriate metrics and k-fold cross-validation, compare their performance with clear tables and visualisations, and discuss limitations and possible future improvements.

Problem, Dataset & Preprocessing Overview

Problem: Binary Classification. The aim is to predict loan default by estimating whether a customer has defaulted based on their application and background features.



Dataset: NBFI Vehicle Loan Repayment Dataset from Hugging Face. It is provided as a CSV file with 1.59 MB size, 8,308 rows, and 36 columns (35 features plus 1 target).



Preprocessing: removed 437 duplicate rows, dropped constant and zero-variance columns, fixed incorrect data types, imputed missing with "Unknown", handled extremes using capping and log, clipping the off-scale value, reduced multicollinearity by dropping, and converted features to numeric form using 0/1 encoding, one-hot encoding.



Feature Engineering and Feature Selection

Feature engineering: We engineered features to make the data easier for models to learn from. This included encoding categorical feature (binary to 0/1, one-hot for nominal, keeping ordinal features as ordered integers) and reducing skew using capping plus log-style transforms on money features. We also fixed a data issue by clipping the off-scale score_2 value.

Feature selection: We removed features that cannot help prediction, such as constant (zero-variance) columns. We also reduced redundancy by dropping nr_children because it was highly correlated with nr_family_members, while keeping credit and loan_annuity since they represent different business meaning. For scale-sensitive models, we used MinMax scaling only on continuous features, not on 0/1 or one-hot columns.

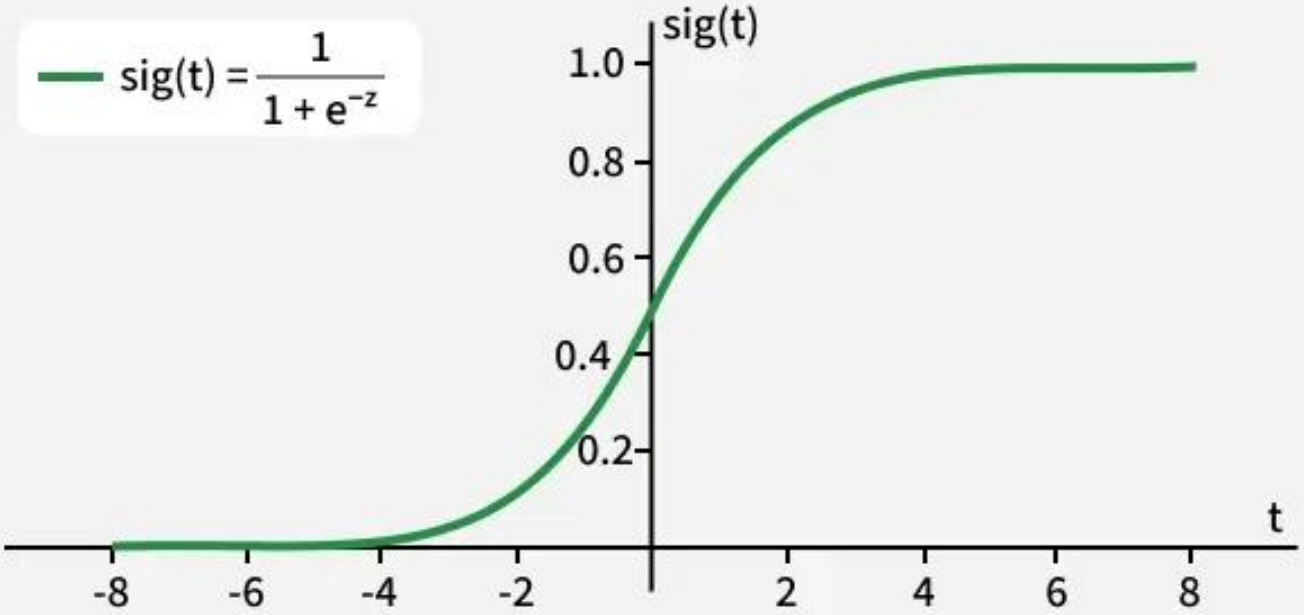
Changes after CA1 Q&A: We changed the hour feature handling. We replaced the cyclical hour_sin and hour_cos encoding with five clear time-of-day categories (before open, before noon, noon break, after noon, after close). This made the feature more interpretable while still capturing time-related behavior.

Model Choices

Although the assignment requires at least four supervised models, we evaluated a broader set because this is a difficult imbalanced classification task. Loan repayment is also influenced by human behavior, which is not fully captured by application features. There is also a data quality risk because some applicants may report details inaccurately or strategically, which adds noise and reduces predictability. For these reasons, we tested a wider range of algorithms to find a model that is more robust to imbalance, noise, and non-linear patterns. Models we used (15 total):

Machine Learning		Deep learning
Logistic Regression	Linear SVM	MLP
Decision Tree	Histogram Gradient Boost	MLP with Dropout
Random Forest	XGBoost	Residual MLP
k-Nearest Neighbours	LightGBM	TabNet
Naive Bayes	CatBoost	FT-Transformer

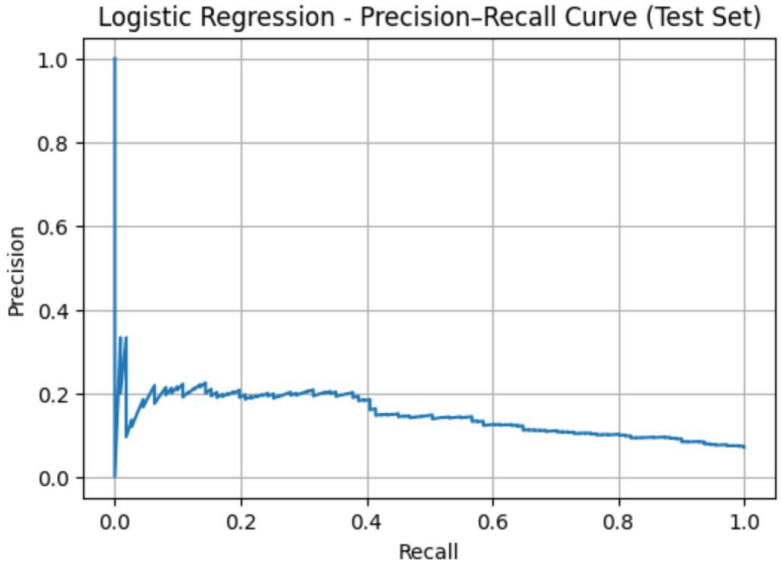
Model 1: Logistic Regression

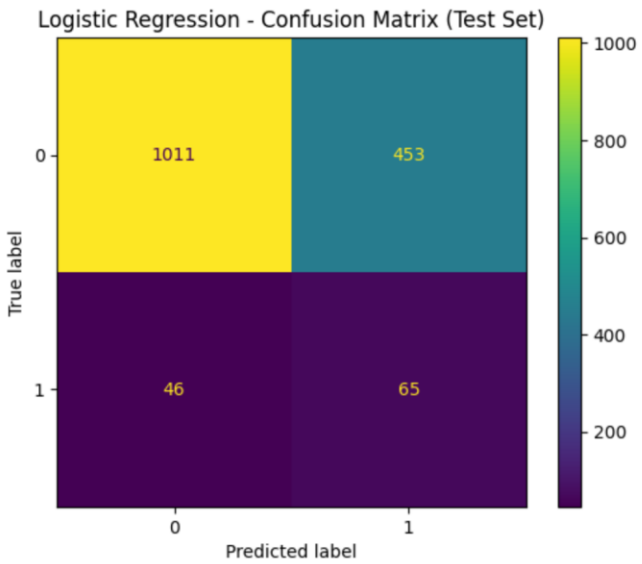


We use Logistic Regression as our first baseline model for predicting default. It is simple, fast, and gives us a clear starting point before trying more complex methods on this imbalanced dataset. It works well as a baseline for binary classification and produces probabilities, which is useful for precision–recall evaluation.

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Model 1: Logistic Regression

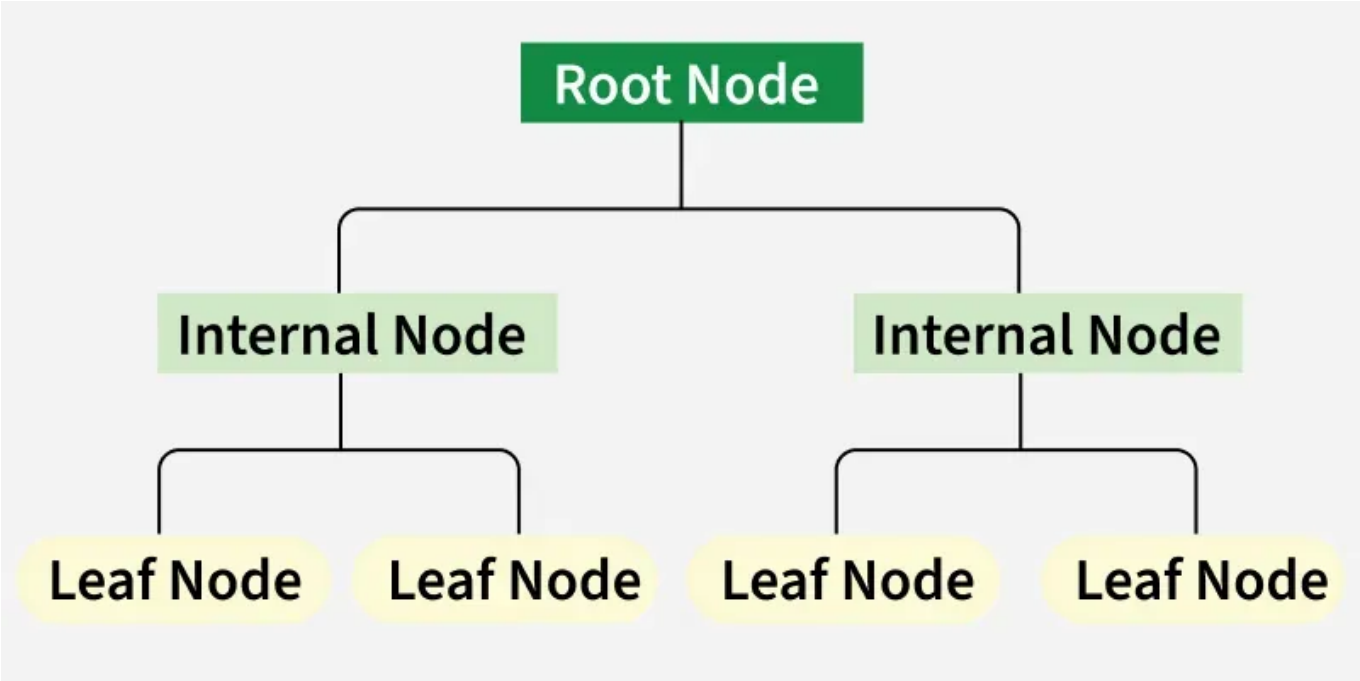




Metric	Mean	Std
PR AUC	0.220787	0.034159
Balanced Accuracy	0.684099	0.011757
Macro F1	0.530194	0.005931

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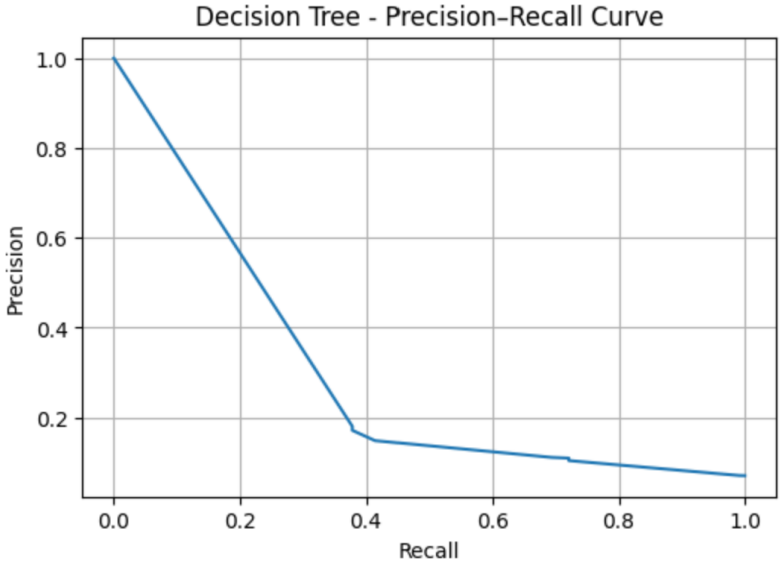
Model 2: Decision Tree

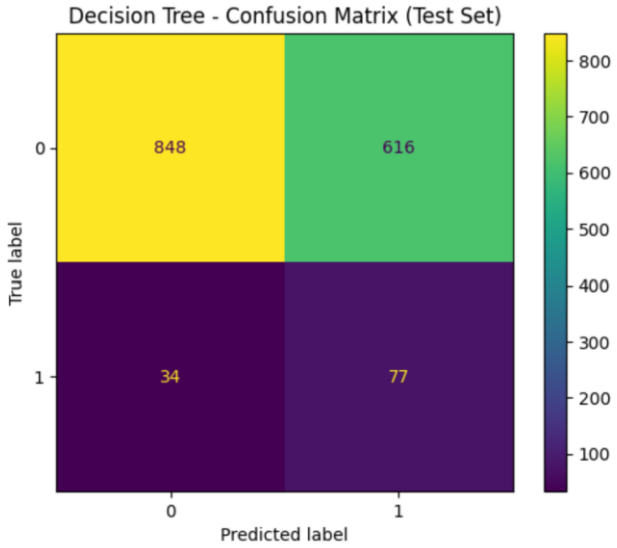


We use a Decision Tree because it can learn non-linear rules and simple if-then patterns. It is a good step after Logistic Regression, especially for tabular data.

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Model 2: Decision Tree

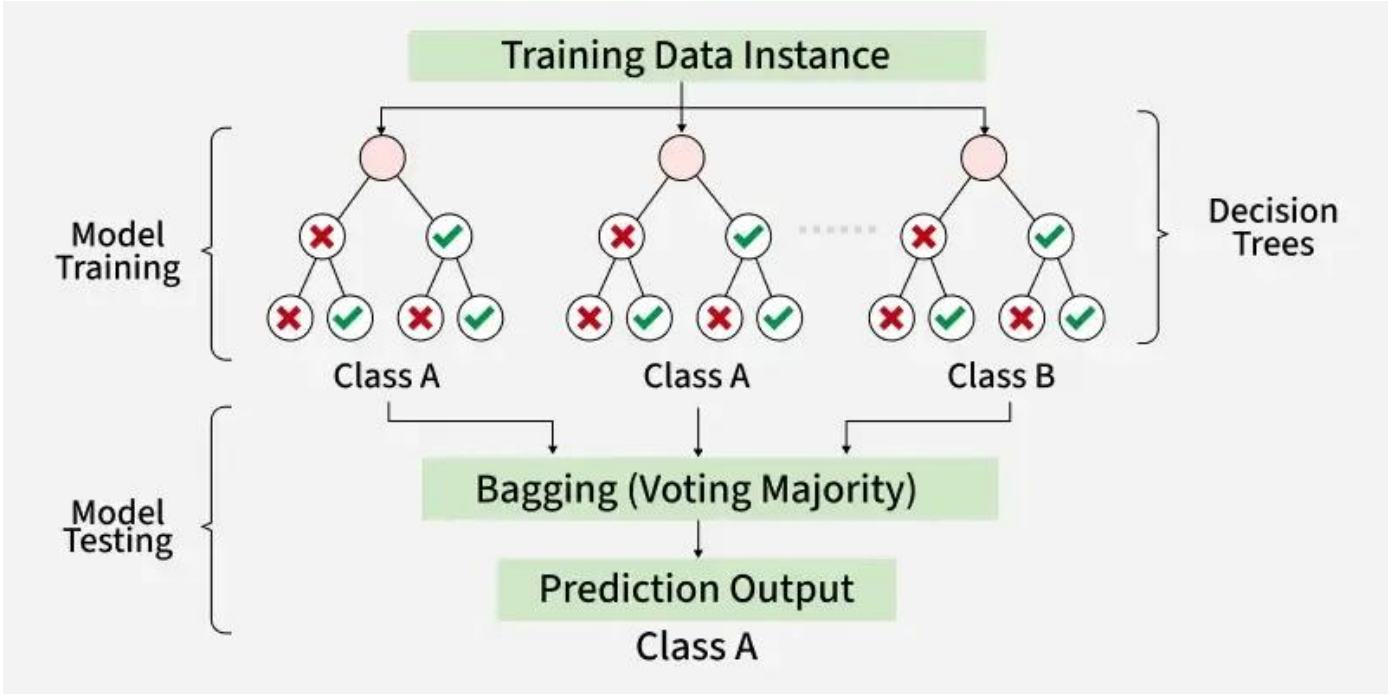




Metric	Mean	Std
PR AUC	0.137858	0.008787
Balanced Accuracy	0.667937	0.010841
Macro F1	0.457918	0.015391

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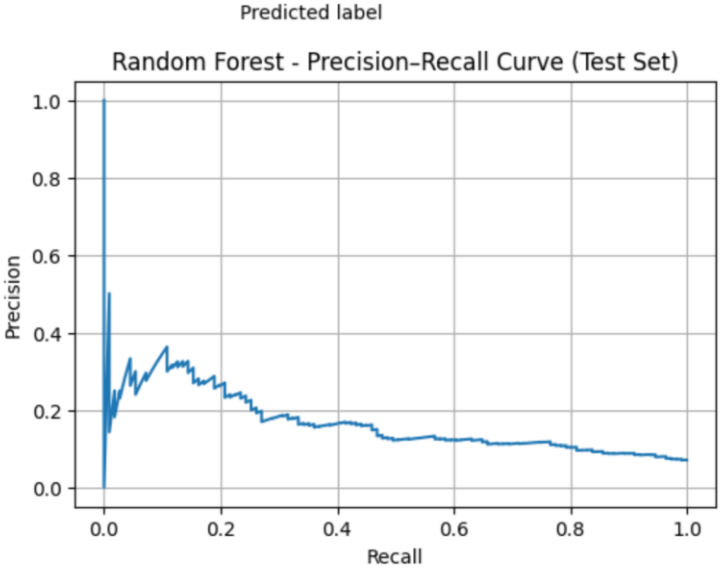
Model 3: Random Forest

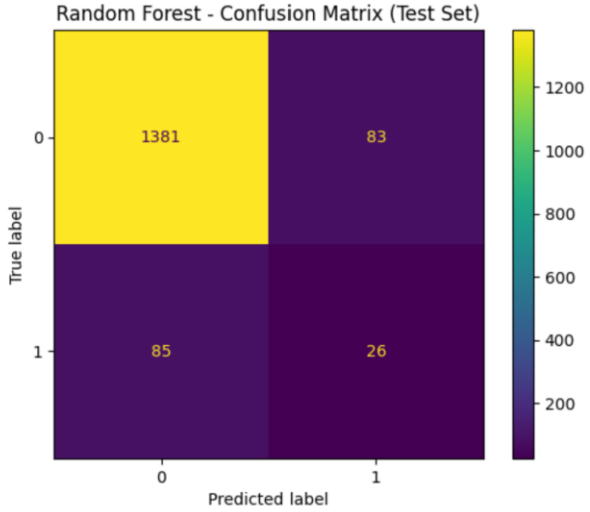


We use Random Forest because it is a strong model for tabular data and it reduces the instability of a single Decision Tree by averaging many trees.

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Model 3: Random Forest

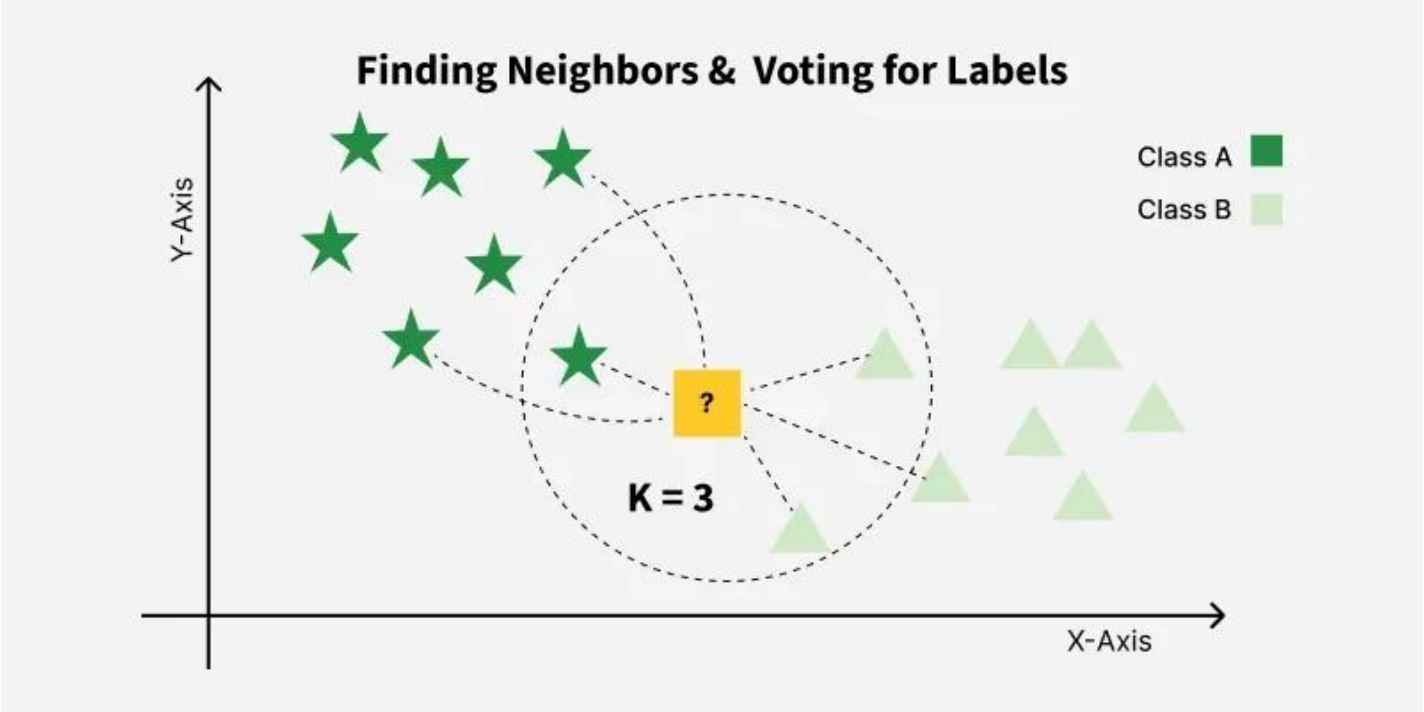




Metric	Mean	Std
PR AUC	0.212925	0.025418
Balanced Accuracy	0.587197	0.014431
Macro F1	0.595913	0.014826

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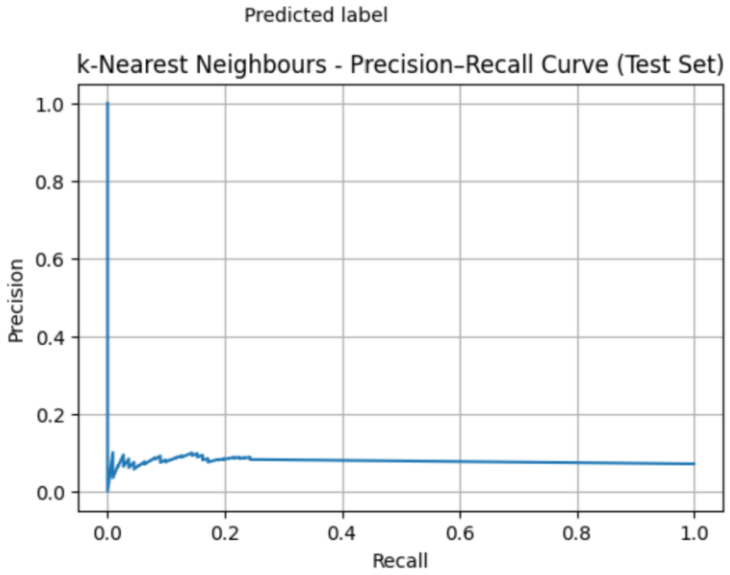
Model 4: k-Nearest Neighbors

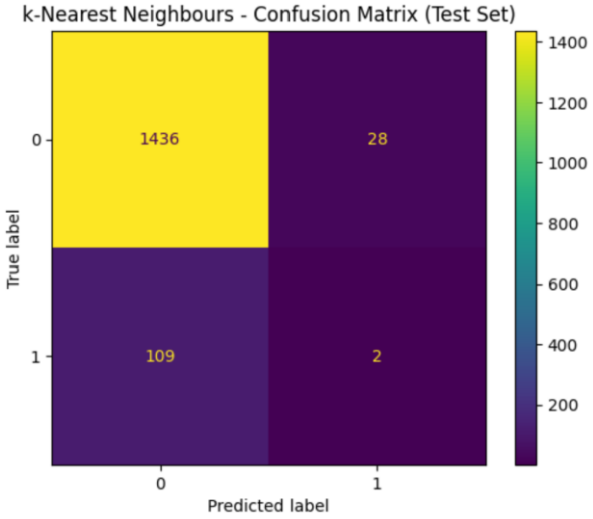


We use k-Nearest Neighbors because it is a simple distance-based baseline that can work well on scaled tabular data, and it gives us a useful comparison against tree models.

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Model 4: k-Nearest Neighbors

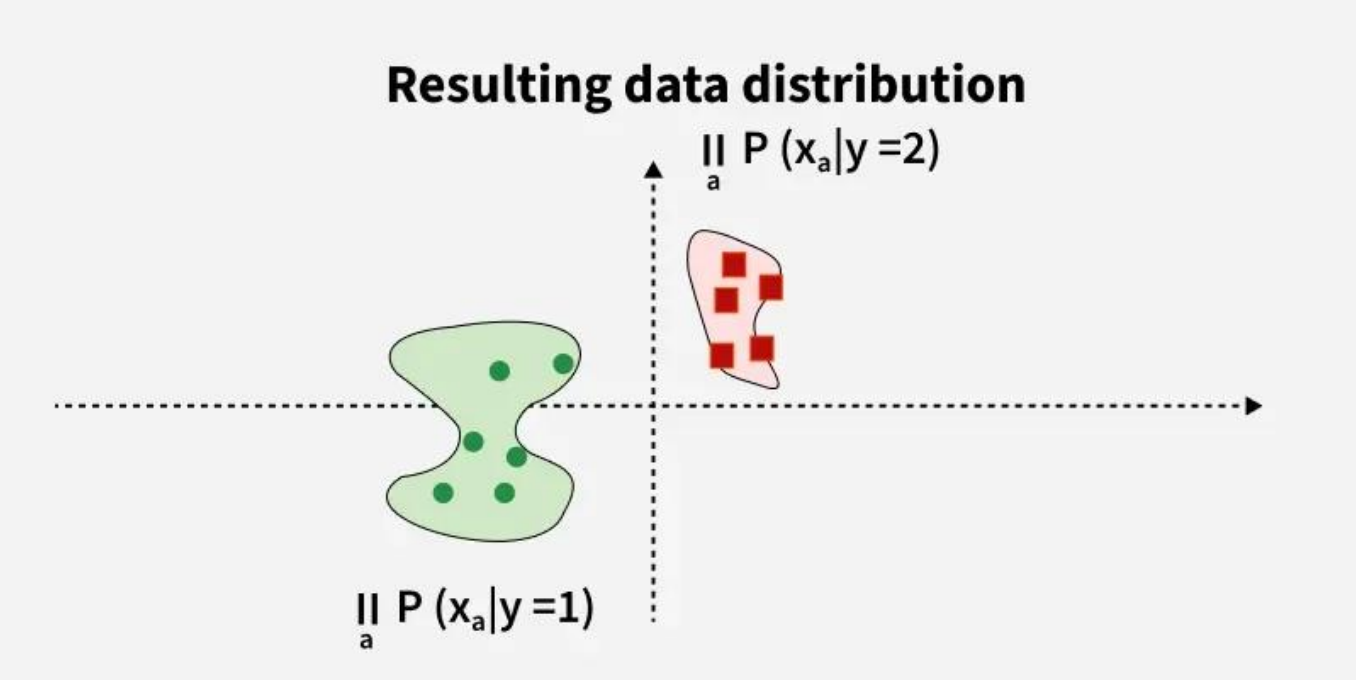




Metric	Mean	Std
PR AUC	0.099428	0.010298
Balanced Accuracy	0.517272	0.011844
Macro F1	0.517582	0.018808

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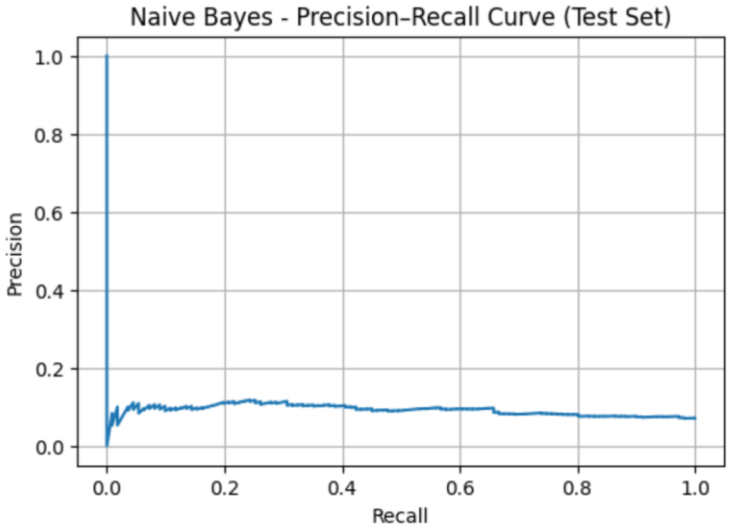
Model 5: Naive Bayes

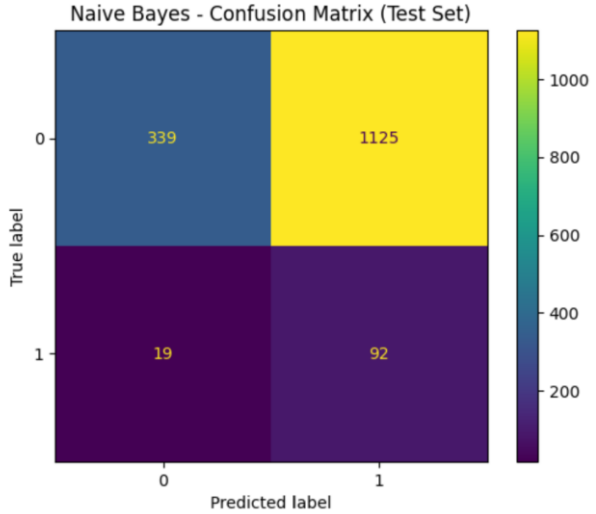


We use Gaussian Naive Bayes as a very fast baseline. It is simple and gives us a quick reference point before moving to more complex models.

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Model 5: Naive Bayes

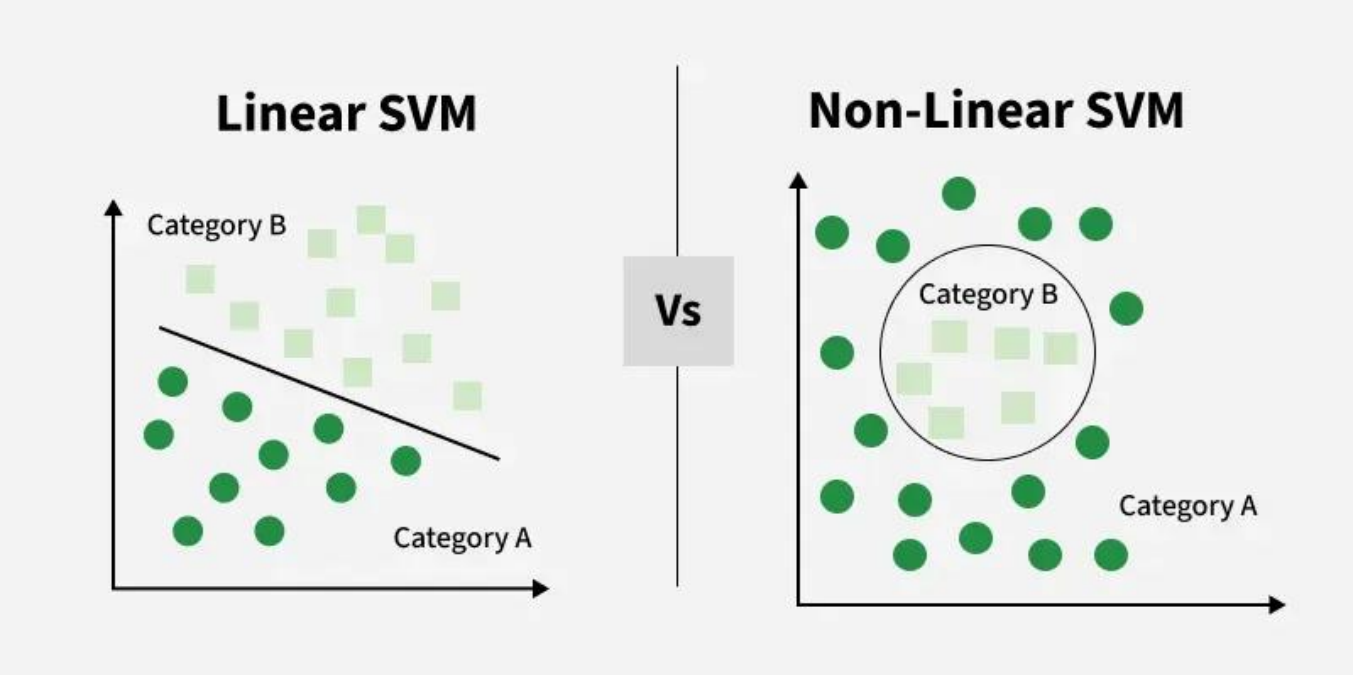




Metric	Mean	Std
PR AUC	0.124086	0.009220
Balanced Accuracy	0.531463	0.021165
Macro F1	0.173803	0.044846

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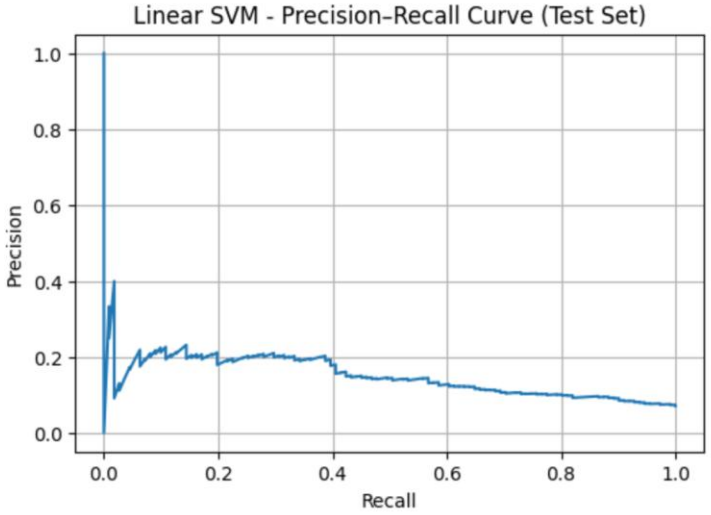
Model 6: Linear SVM

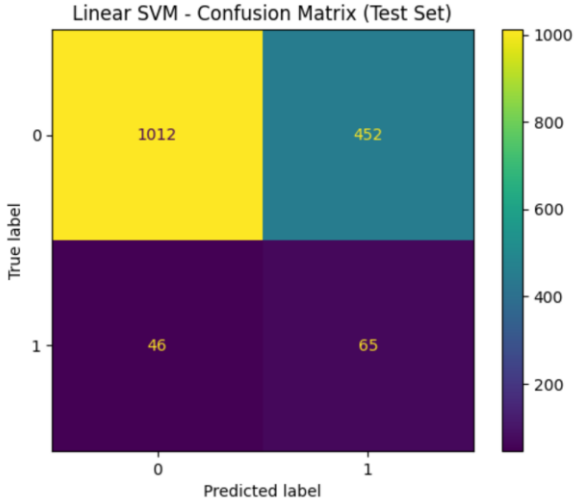


We use a Linear SVM because it often works well on high-dimensional tabular data, especially after one-hot encoding. It is also a good comparison to Logistic Regression because both are linear, but SVM can behave differently with the same features.

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Model 6: Linear SVM

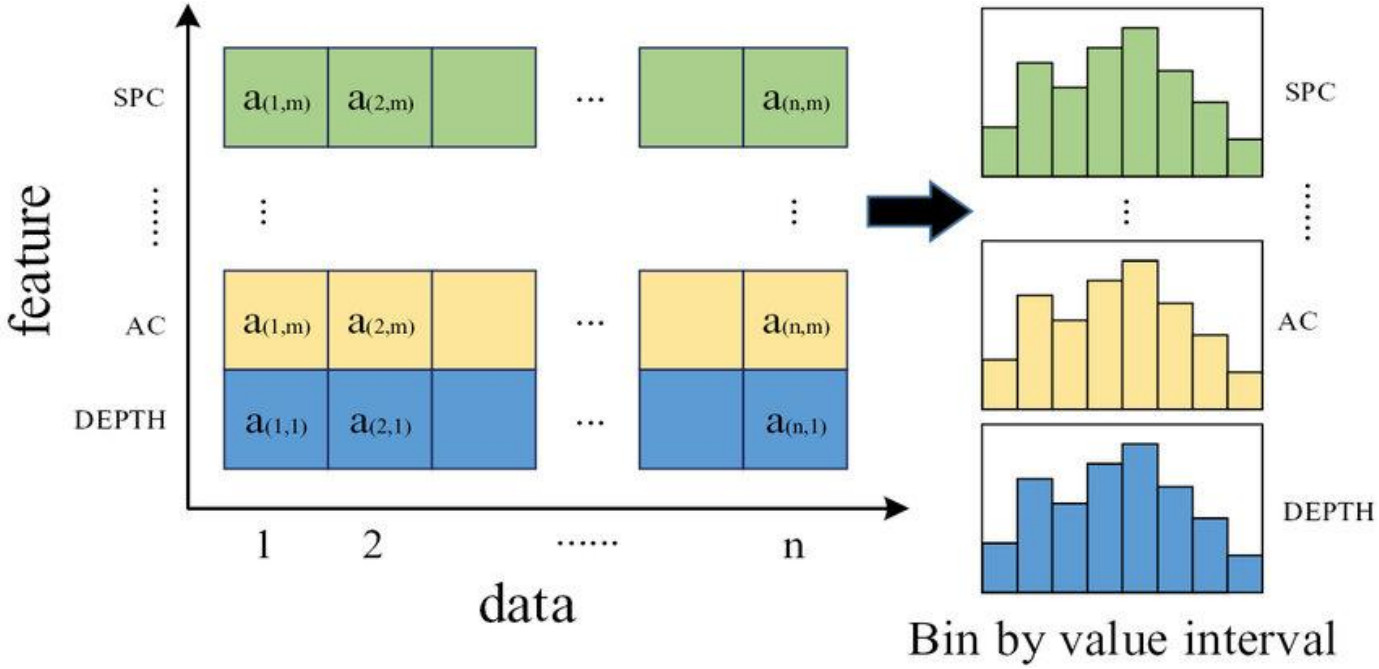




Metric	Mean	Std
PR AUC	0.221918	0.034482
Balanced Accuracy	0.685662	0.012821
Macro F1	0.530131	0.005562

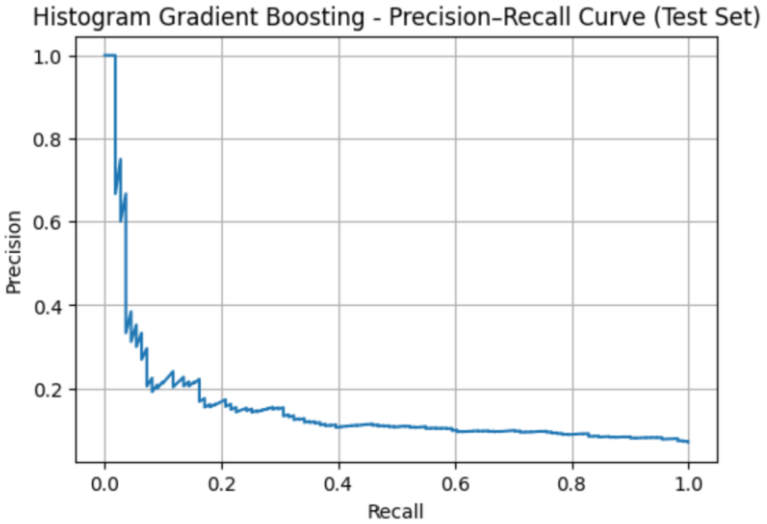
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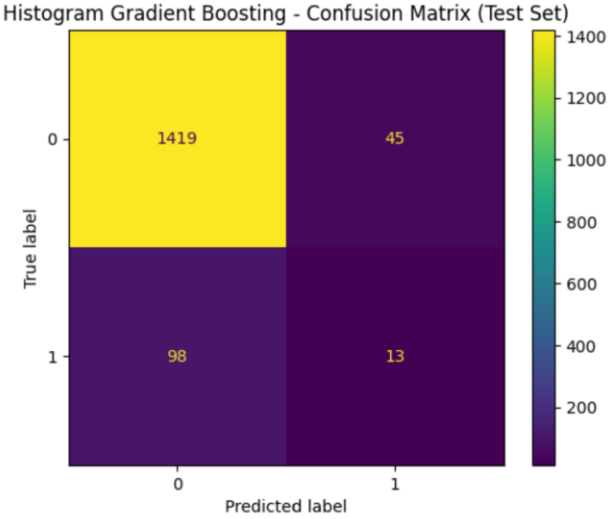
Model 7: Histogram Gradient Boost



We use Histogram Gradient Boosting because it is usually strong on tabular data and can learn non-linear patterns and feature interactions.

Model 7: Histogram Gradient Boost

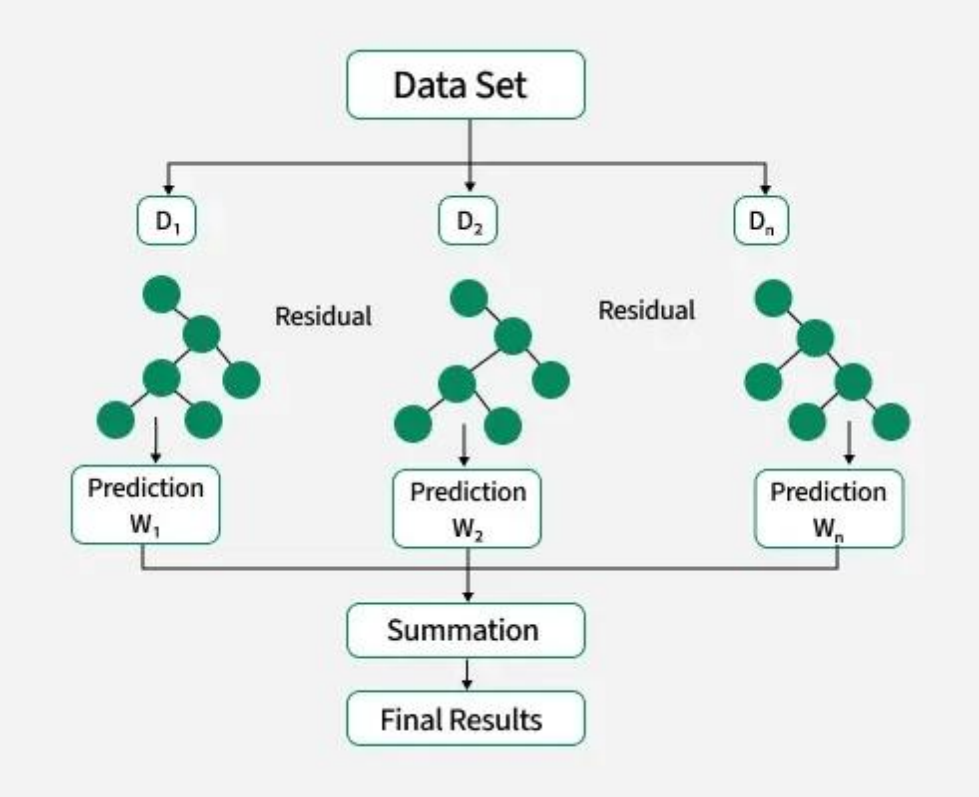




Metric	Mean	Std
PR AUC	0.191064	0.040129
Balanced Accuracy	0.542725	0.016282
Macro F1	0.554611	0.021606

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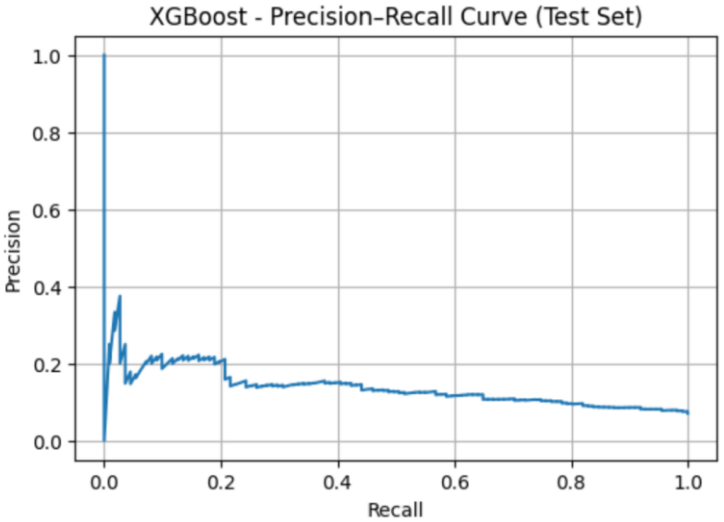
Model 8: XGBoost

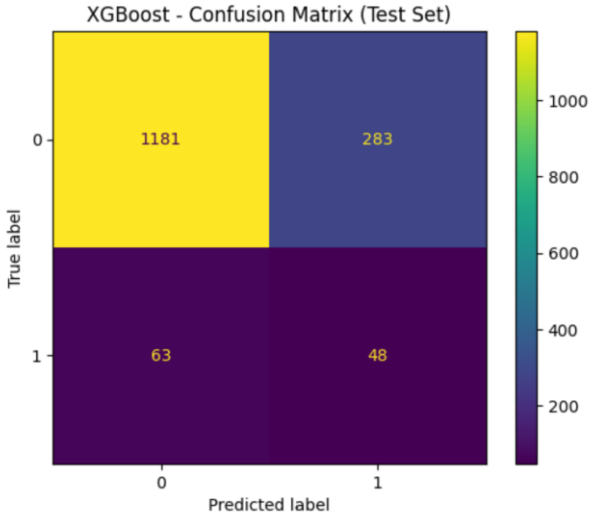


We use XGBoost because it is usually one of the strongest choices for tabular data and it can learn complex non-linear patterns and feature interactions. It is very flexible through tuning. It can overfit if the model is too complex, and it needs careful tuning to balance performance and generalisation.

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Model 8: XGBoost

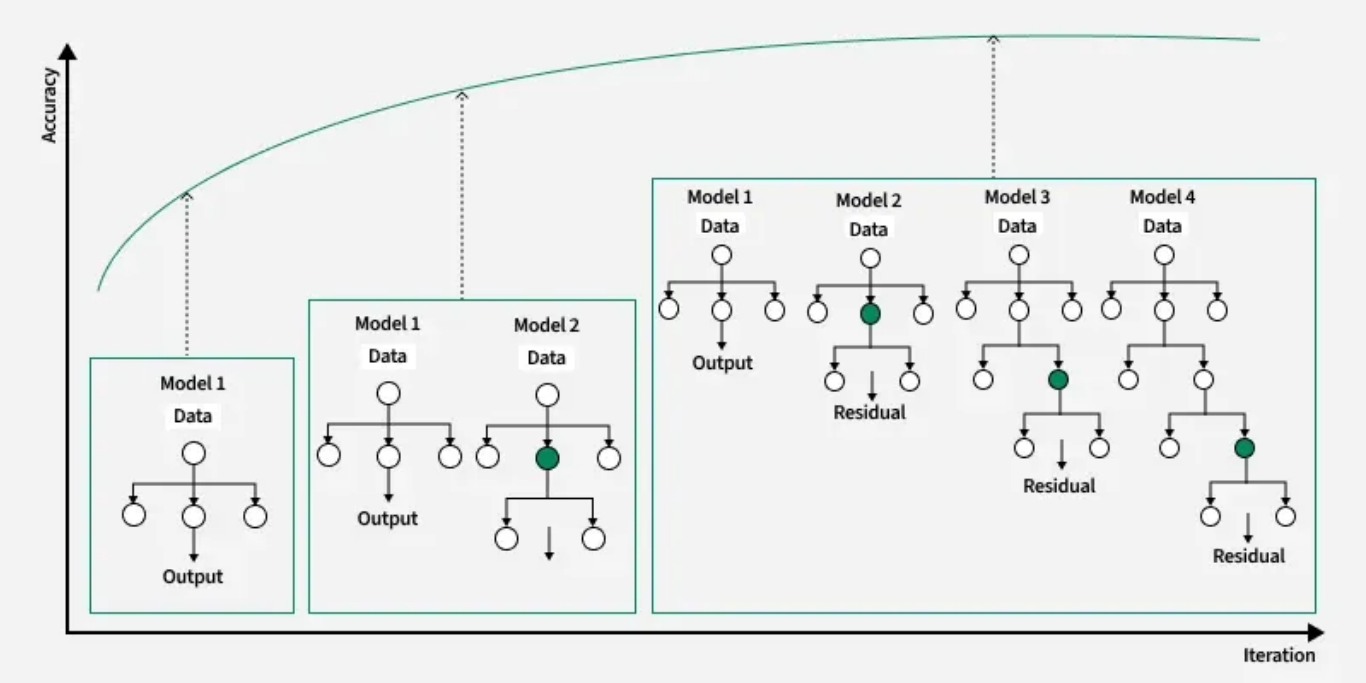




Metric	Mean	Std
PR AUC	0.198011	0.014241
Balanced Accuracy	0.661698	0.013504
Macro F1	0.579506	0.007847

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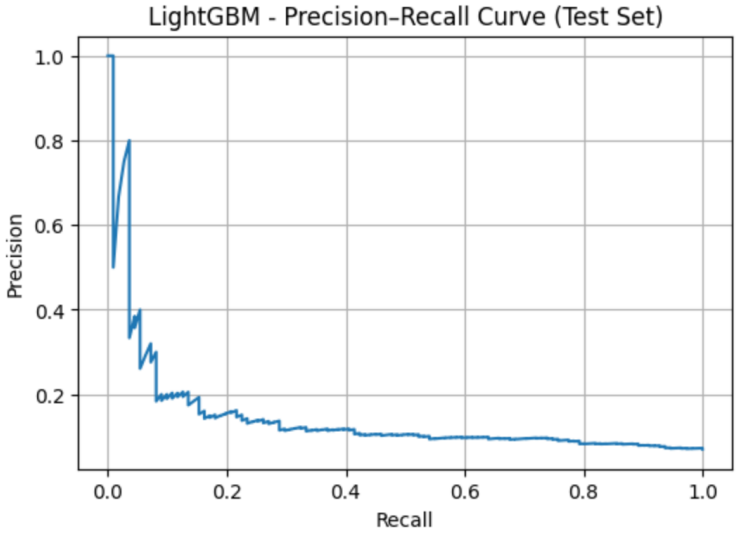
Model 9: LihgtGBM

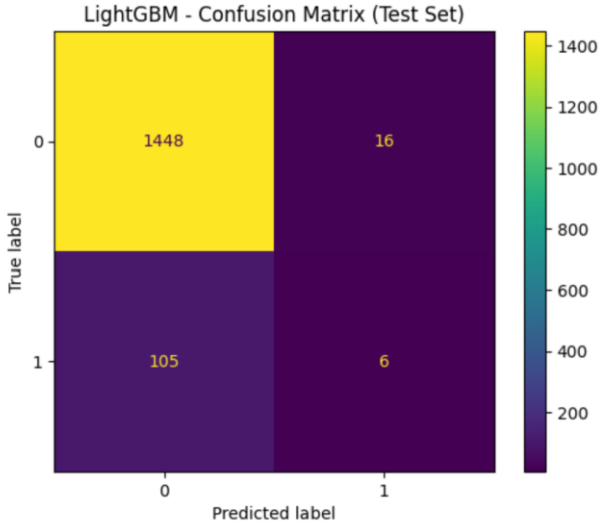


We use LightGBM because it is usually a strong choice for tabular data and it can learn non-linear patterns and feature interactions. It often performs well on structured credit style data, handles complex relationships, and does not need feature scaling.

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Model 9: LihgtGBM

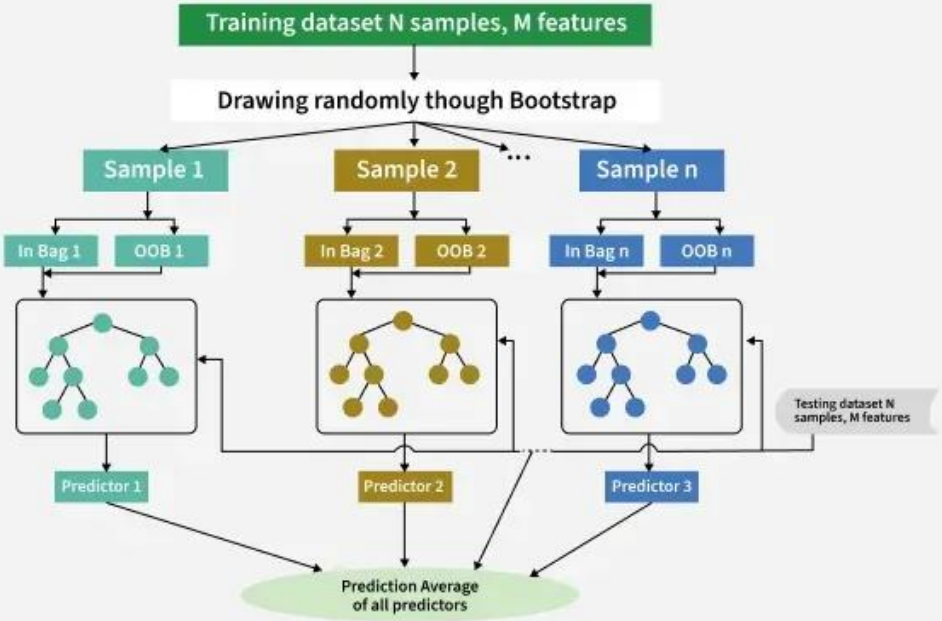




Metric	Mean	Std
PR AUC	0.203915	0.039115
Balanced Accuracy	0.541196	0.013433
Macro F1	0.556152	0.020862

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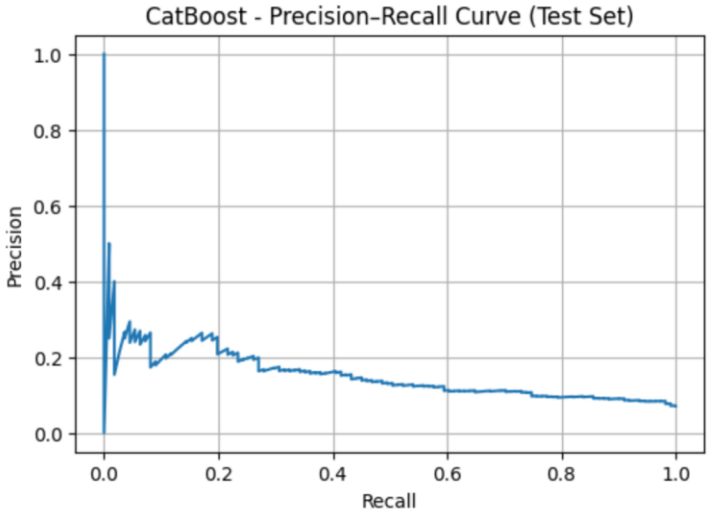
Model 10: CatBoost

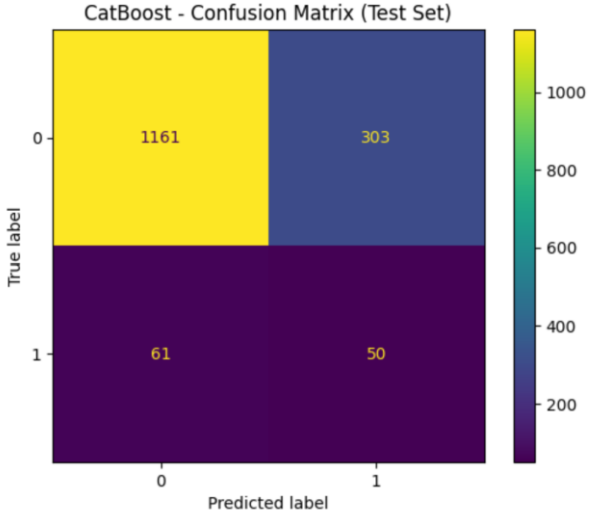


We use CatBoost as another strong boosting model for tabular data. It is good at learning non-linear patterns and feature interactions, which can matter for default prediction. It often performs strongly on structured data, handles complex relationships, and usually works well without heavy feature engineering.

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Model 10: CatBoost

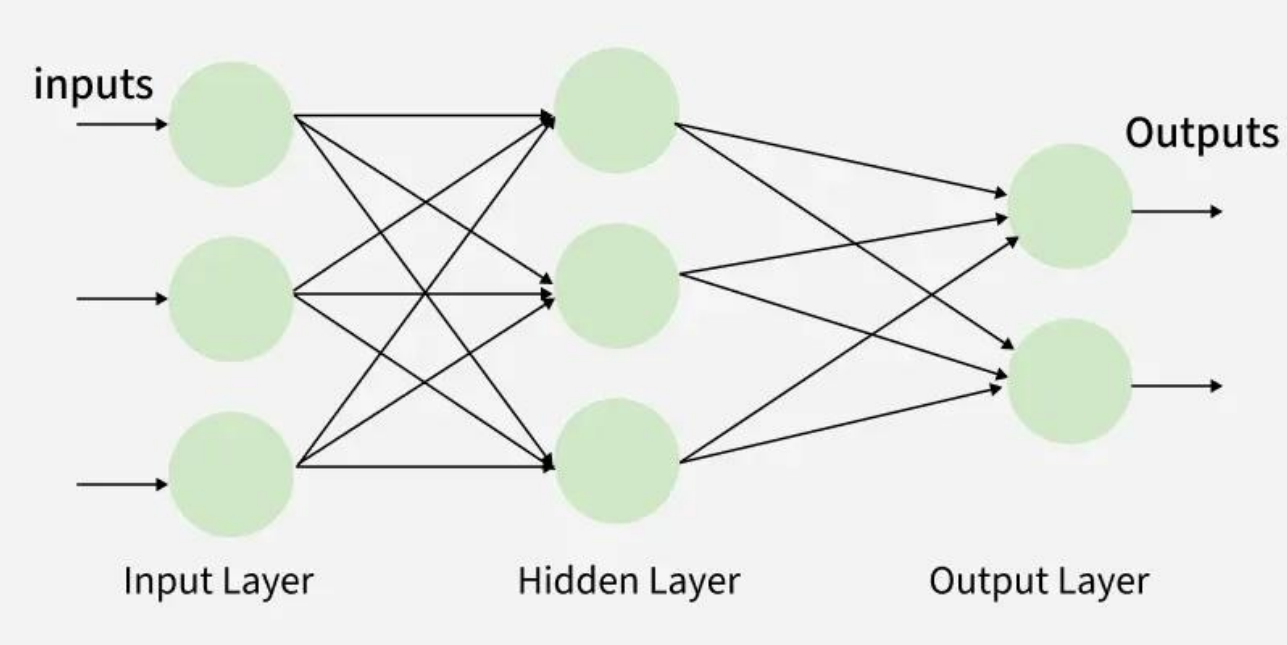




Metric	Mean	Std
PR AUC	0.209338	0.021271
Balanced Accuracy	0.668478	0.015139
Macro F1	0.579426	0.008379

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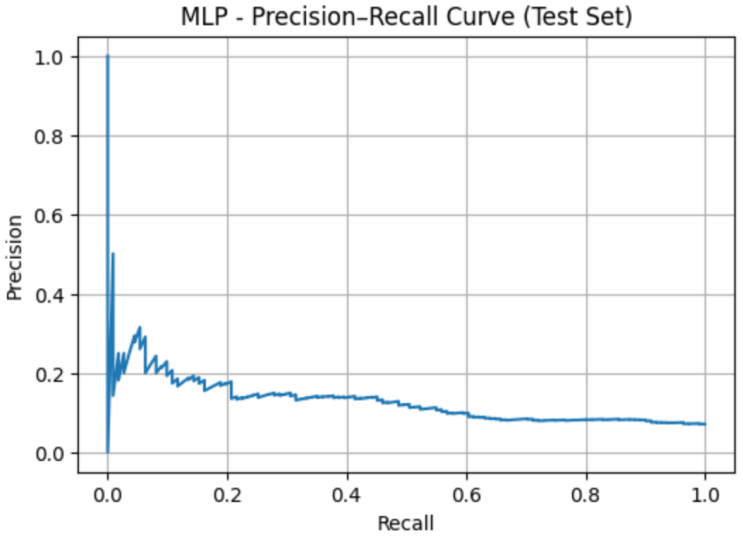
Model 11: MLP

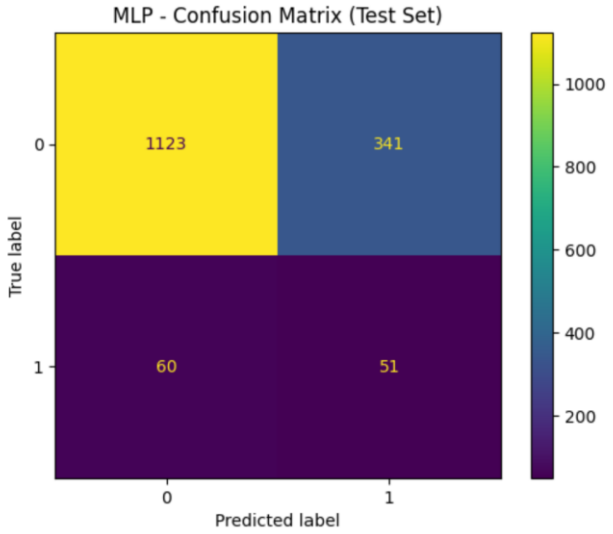


We use a simple deep learning MLP as our first neural network model for this task. It can learn non-linear patterns in tabular data and gives us a clear deep learning baseline to compare against the traditional models. It can capture complex relationships between features, it works well with many input columns, and it outputs probabilities which we evaluate using precision–recall area under the curve.

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Model 11: MLP

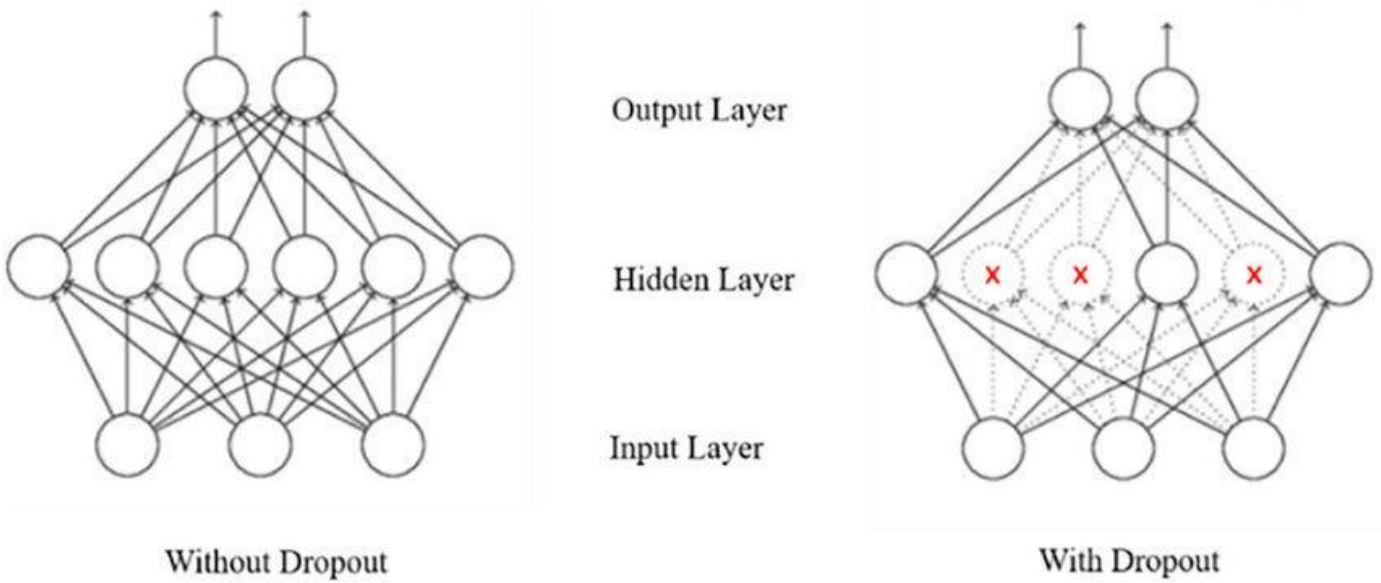




Metric	Mean	Std
PR AUC	0.223033	0.030050
Balanced Accuracy	0.670831	0.019015
Macro F1	0.556531	0.026538

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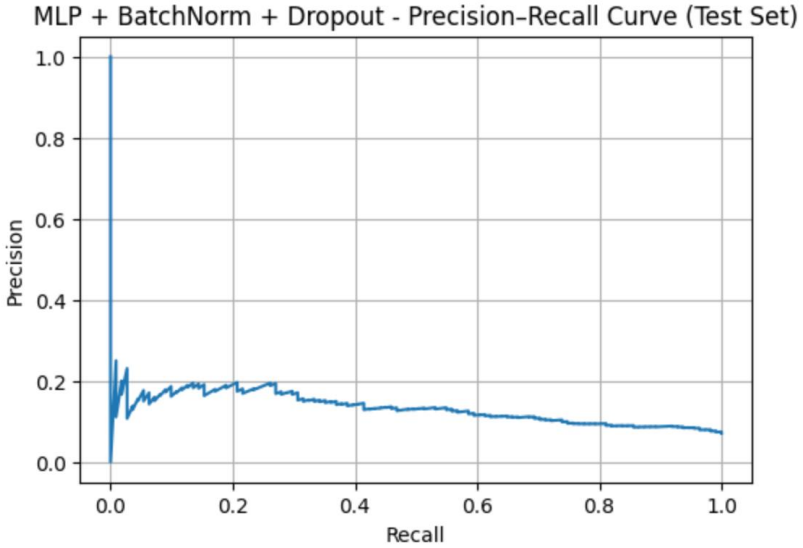
Model 12: MLP with Dropout

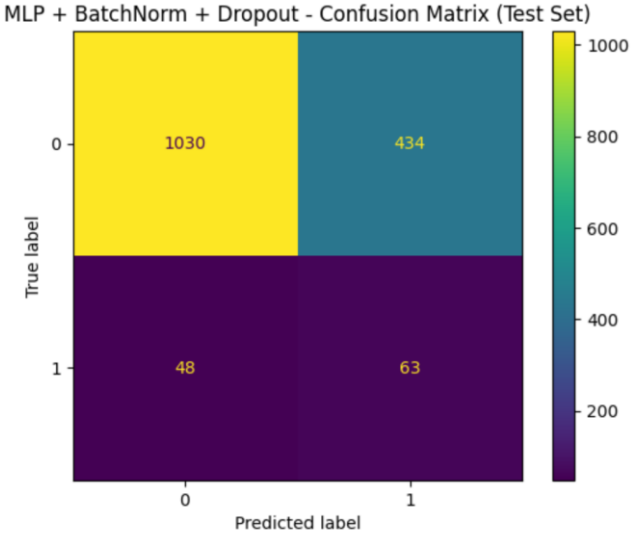


We use this neural network to capture non-linear patterns in the data. Batch normalization helps training stay stable, and dropout helps generalization. It can model complex relationships between features, and it produces probabilities that work well with precision–recall evaluation on an imbalanced target.

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Model 12: MLP with Dropout

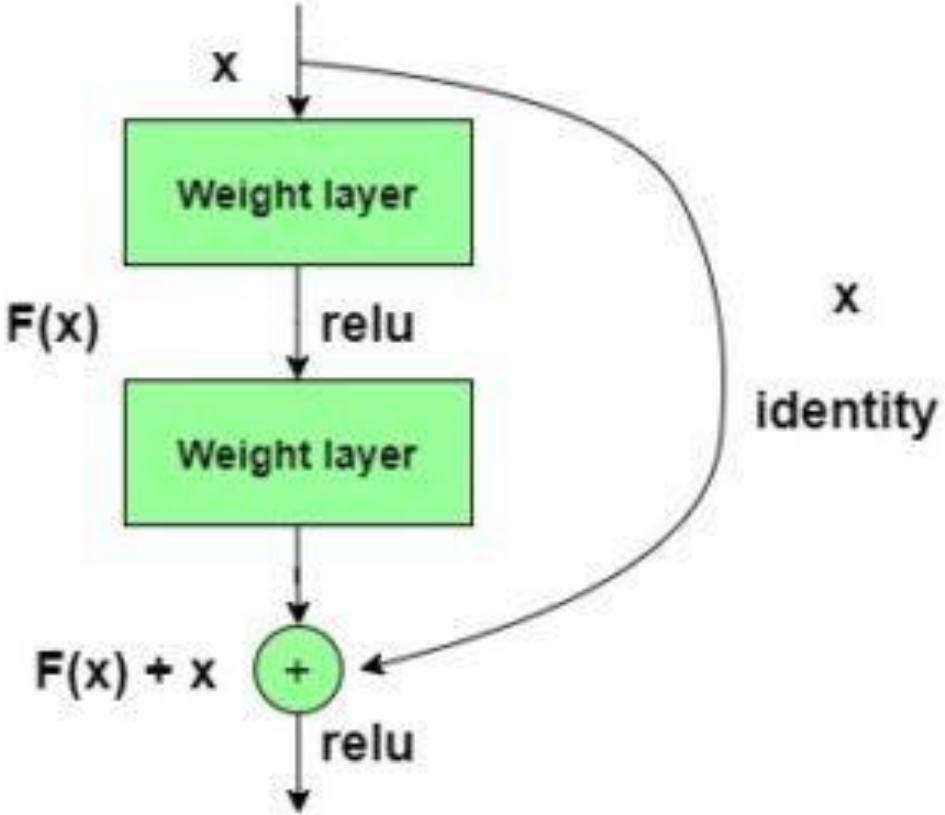




Metric	Mean	Std
PR AUC	0.223258	0.038381
Balanced Accuracy	0.678035	0.010005
Macro F1	0.542768	0.005793

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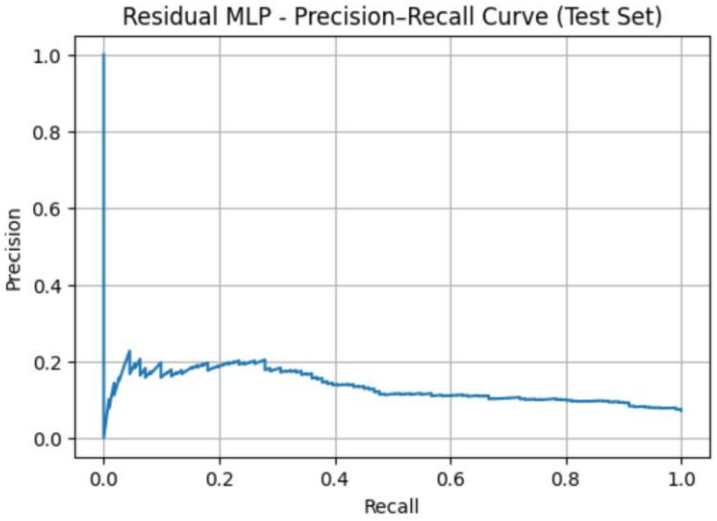
Model 13: Residual MLP

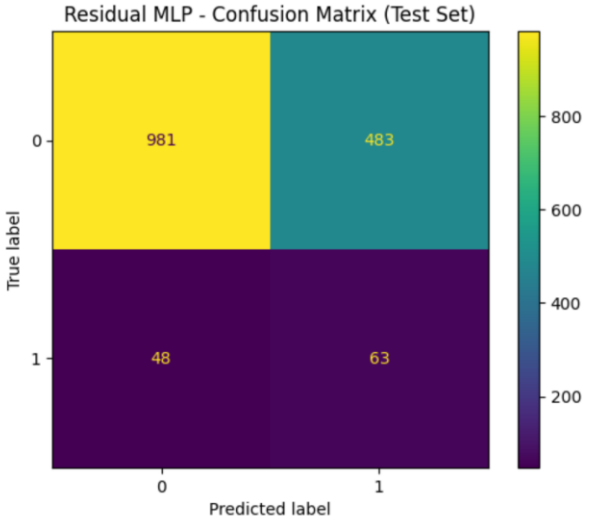


We use a Residual MLP to handle non-linear patterns in tabular data. The skip connections help training stay stable as the network gets deeper. It can learn complex feature interactions, and the residual blocks help gradients flow better than a plain MLP. It outputs probabilities, so it fits our precision–recall based evaluation.

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Model 13: Residual MLP

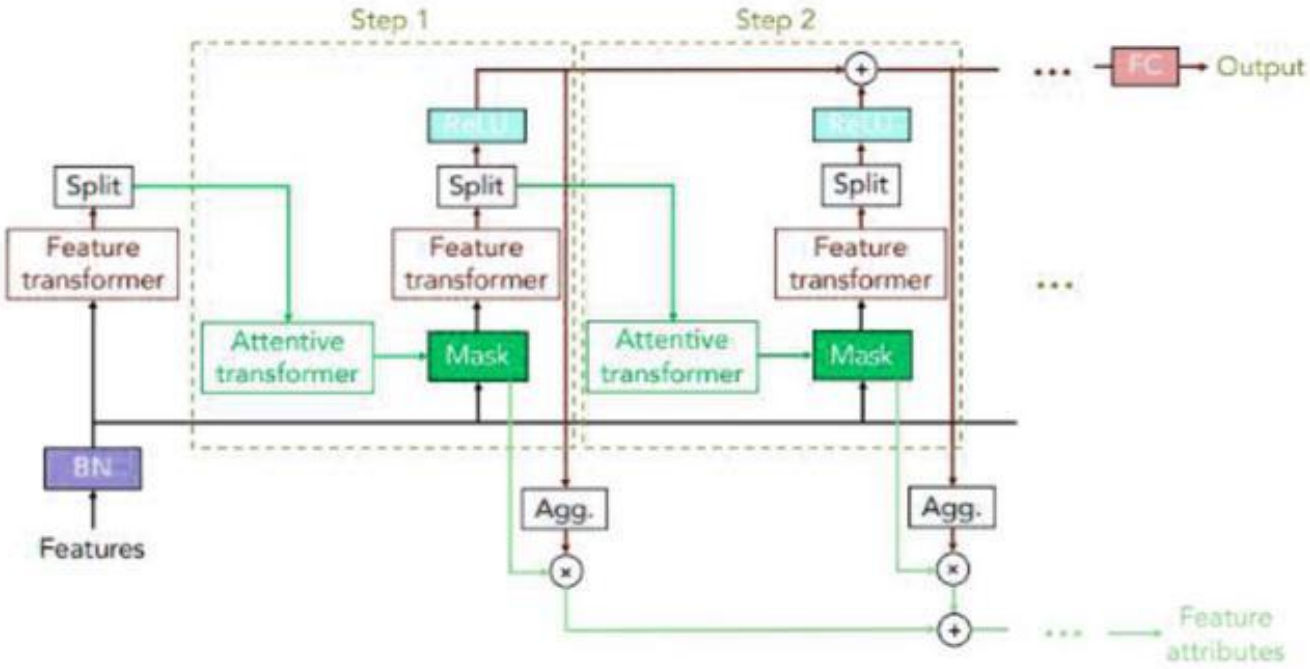




Metric	Mean	Std
PR AUC	0.218562	0.021930
Balanced Accuracy	0.671059	0.024331
Macro F1	0.491813	0.025328

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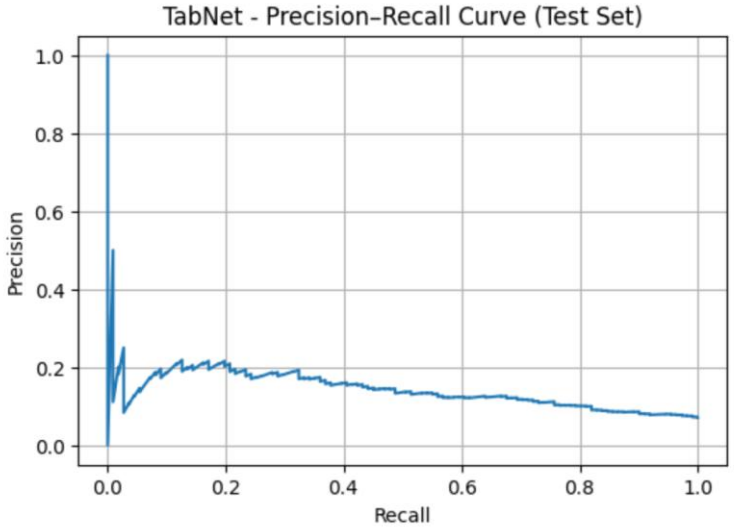
Model 14: TabNet

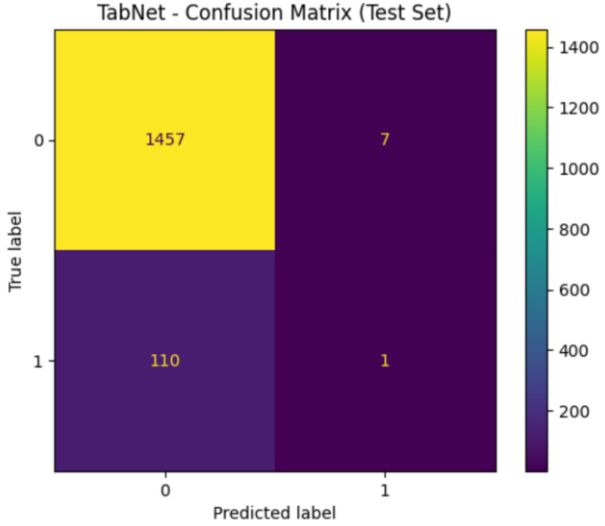


We used TabNet as a deep learning model designed for tabular data. It can learn non-linear patterns and feature interactions, and it also has built-in feature selection behaviour through its attention masks. It can model complex relationships without manual interaction features.

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Model 14: TabNet

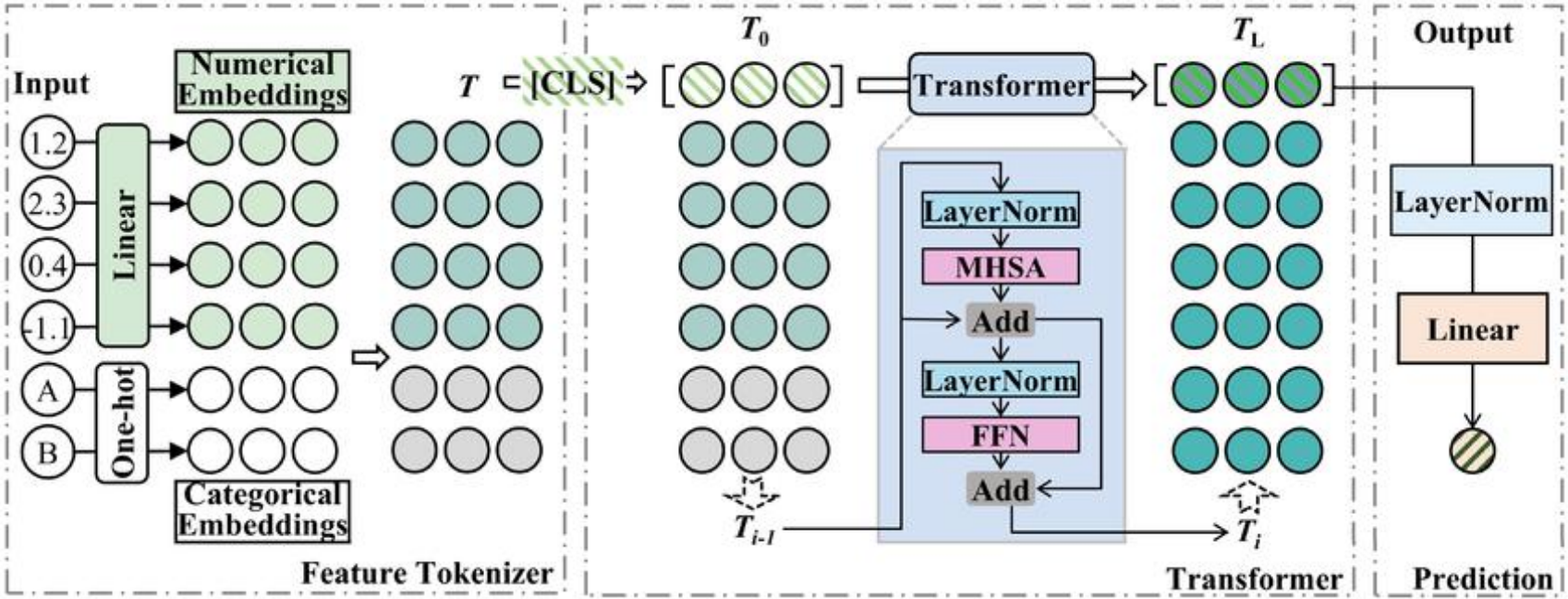




Metric	Mean	Std
PR AUC	0.135065	0.033169
Balanced Accuracy	0.524019	0.020569
Macro F1	0.525392	0.027718

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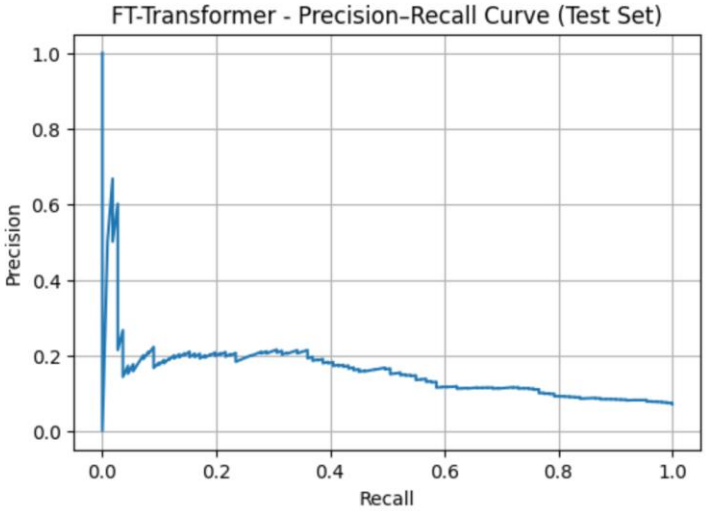
Model 15: FT-Transformer

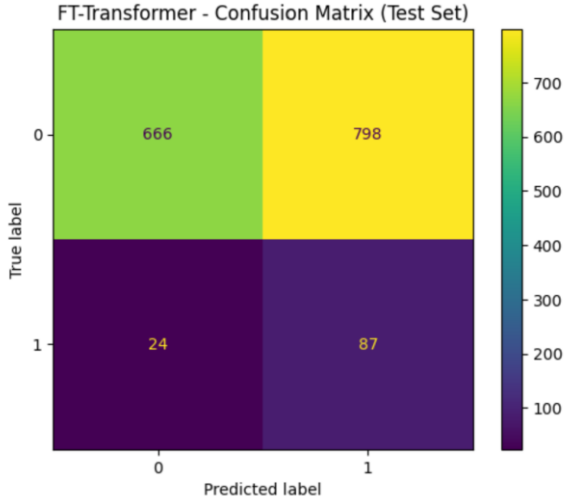


We used FT-Transformer because it is designed for tabular data and can learn complex feature interactions, which can matter when default behavior is difficult to predict. It is also a relatively new deep learning approach based on the Transformer idea, which is the same general family of models used in modern language models such as ChatGPT, but applied here to structured tabular features.

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Model 15: FT-Transformer





Metric	Mean	Std
PR AUC	0.194310	0.008409
Balanced Accuracy	0.621532	0.049784
Macro F1	0.452639	0.173782

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Evaluation

Evaluation method: We use **5-fold Cross-Validation** on the combined train and validation data for each model. We report a small summary table with the mean and standard deviation for key metrics.

Key metrics: We treat this as an imbalanced classification problem, so we do not rely on plain accuracy.

Instead we report:

- **PR AUC (Average Precision)** as the main metric. It focuses on how well we rank and detect the rare default class, and it is much more informative than accuracy when positives are scarce.
- **Balanced Accuracy** to measure performance fairly across both classes. It avoids the majority class dominating the score.
- **Macro F1** to balance precision and recall while giving equal importance to both classes, which is useful when the minority class matters most.

Insights

Most stable model (lowest CV PR AUC std):

Most stable (lowest CV PR AUC std):

	model	cv_pr_auc_mean	cv_pr_auc_std
9	FT-Transformer	0.194310	0.008409
11	Decision Tree	0.137858	0.008787
13	Naive Bayes	0.124086	0.009220
14	k-Nearest Neighbours	0.099428	0.010298
8	XGBoost	0.198011	0.014241

Biggest CV and test drop (possible overfit):

Largest CV → test PR AUC drop:

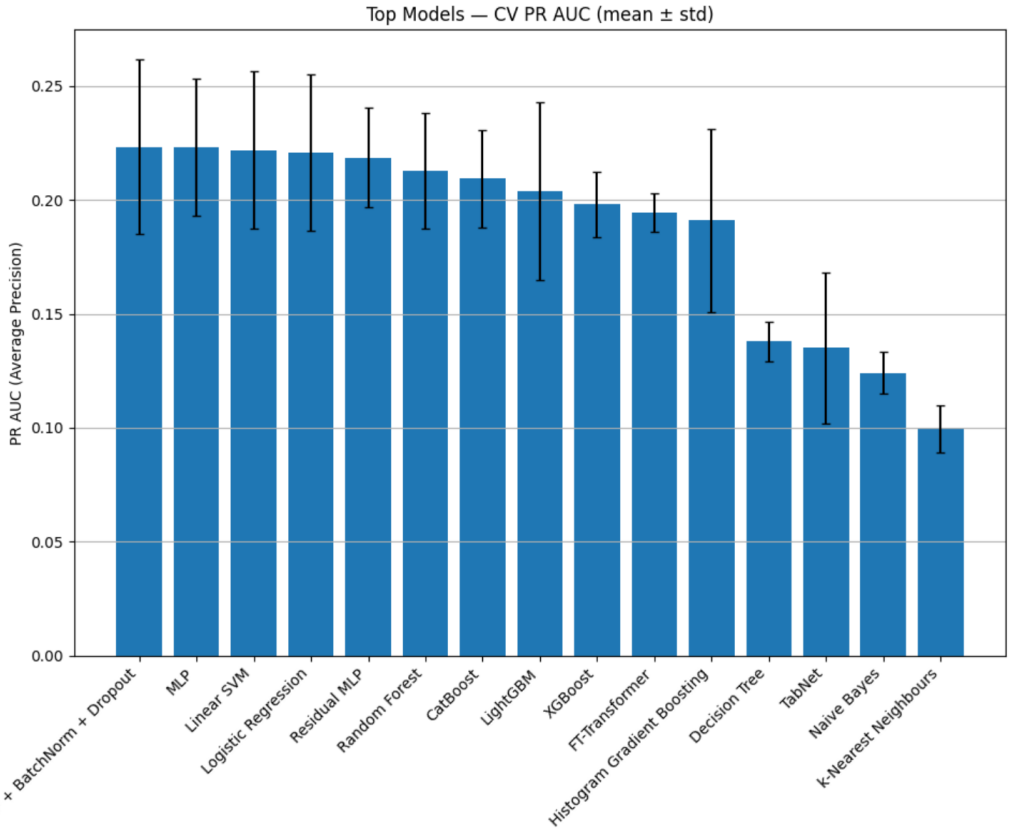
	model	cv_pr_auc_mean	test_pr_auc	cv_to_test_pr_auc_drop
1	MLP	0.223033	0.131191	0.091843
0	MLP + BatchNorm + Dropout	0.223258	0.134881	0.088377
4	Residual MLP	0.218562	0.134843	0.083719
2	Linear SVM	0.221918	0.151649	0.070269
3	Logistic Regression	0.220787	0.150866	0.069921

Multi-metric score (we just sum 3 metrics):

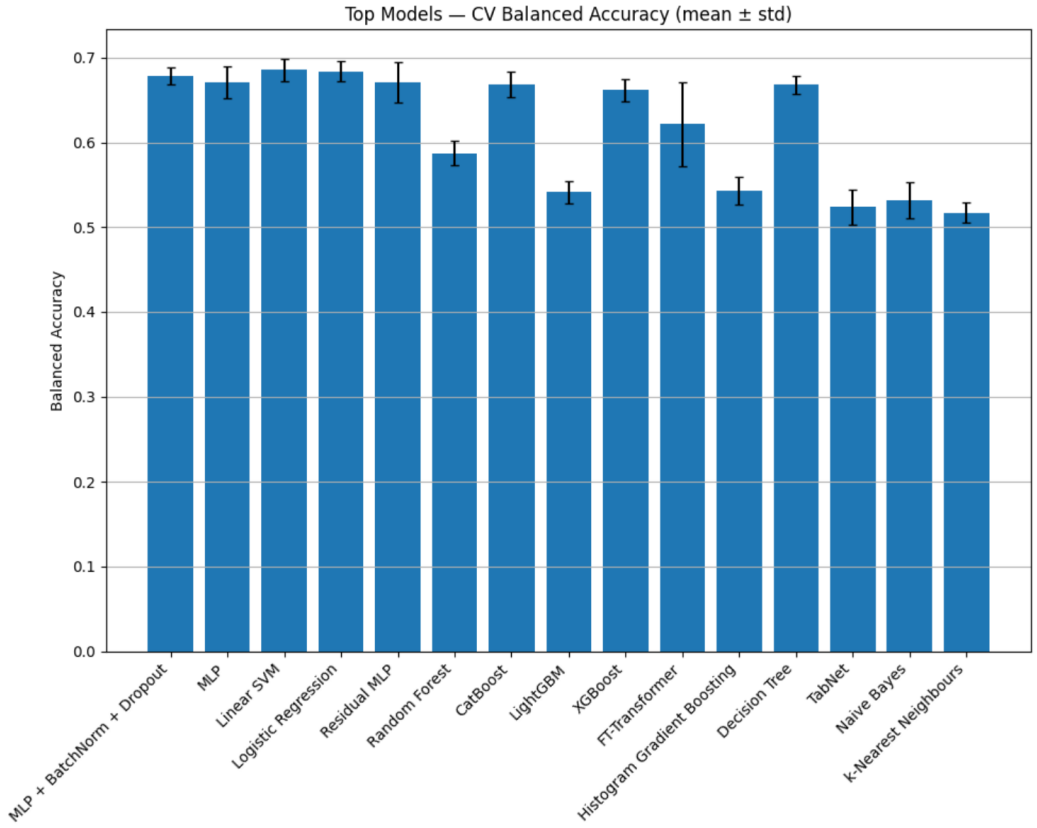
Best overall by simple rank-sum (PR AUC + Balanced Accuracy + Macro F1):

	model	cv_pr_auc_mean	cv_balanced_accuracy_mean	cv_f1_macro_mean	rank_sum
0	MLP + BatchNorm + Dropout	0.223258	0.678035	0.542768	11.0
1	MLP	0.223033	0.670831	0.556531	11.0
2	Linear SVM	0.221918	0.685662	0.530131	13.0
3	Logistic Regression	0.220787	0.684099	0.530194	14.0
4	CatBoost	0.209338	0.668478	0.579426	16.0
5	Random Forest	0.212925	0.587197	0.595913	17.0

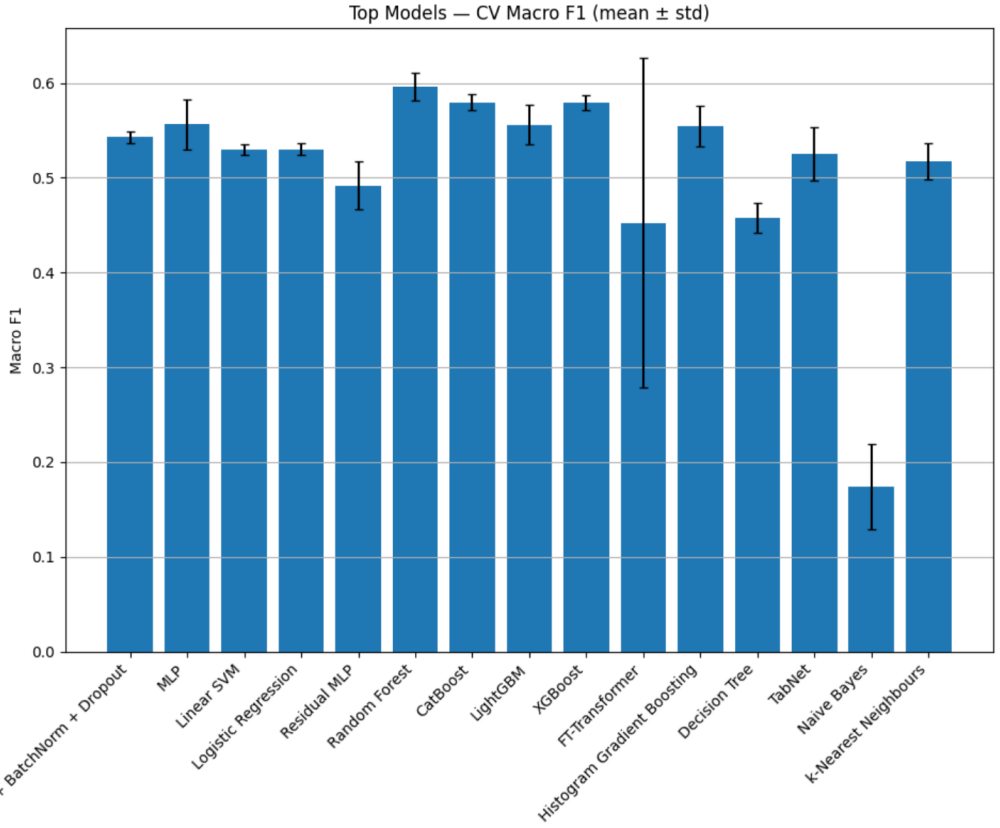
Visual comparision CV PR AUC



Visual comparision CV Balanced Accuracy



Visual comparision CV Macro F1



Final choice

```
Best CV PR AUC: MLP + BatchNorm + Dropout 0.2232576543688325 5
Best CV Balanced Accuracy: Linear SVM 0.6856618430797803
Best CV Macro F1: Random Forest 0.5959125772808445
Most stable (lowest CV PR AUC std): FT-Transformer 0.008409247481279292
```



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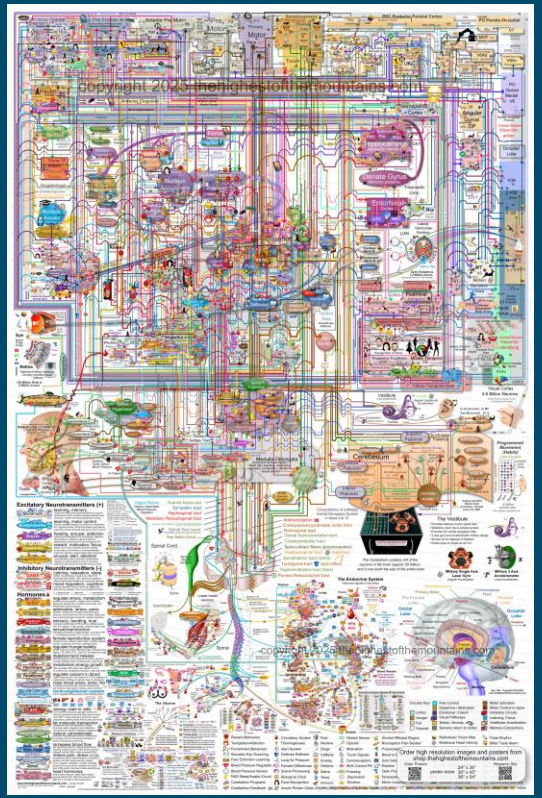
Future work and model improvements

We list a few things we can try next to improve the results :

- Do better **hyperparameter search**, but only for the best models.
- Use **automatic feature selection** or **dimensionality reduction** to test which features really help and which ones add noise.
- Try an **models ensemble**, for example combine the best boosting model with the best deep learning model.
- Optimize training for **GPU**.
- Optimize training for **TPU**. It can calculate large matrices very fast, so we can train deep models quicker, try more tuning, and models like FT-Transformer may give better performance with enough training.
- Keep searching for **new models** and **new methods**.

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Model 16: Human Brain Neural Network



Trained for a few million years on dataset called LIFE

Energy-efficient, needs only 20W a day (one pizza)

Weighs roughly 1300 gram

Two redundant processors (left and right hemisphere)

Nightly updates SLEEP

Tendency to overfit

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